

Chinnaswamy Purra

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PROFESSIONAL EXPERIENCE

Feb 2026 – Aug 2026
Wolfsburg, Germany

Volkswagen AG

Master Thesis Researcher

- **Developed a compact multimodal GUI-agent for edge-deployed automotive infotainment navigation** and curated a leakage-safe CUPRA dataset comprising 248 task trajectories and 1,026 annotated UI steps across six interaction classes.
- **Built an end-to-end structured action pipeline** and **fine-tuned Qwen3-VL-2B-Instruct** using LoRA and reasoning-oriented supervision, boosting held-out **action accuracy** from 38.3% to 73.9% and **joint action-grounding** from 0.9% to 40.0%.
- **Implemented GRPO reward-guided refinement** with custom **action-grounding** rewards, achieving 80.0% action accuracy and 0.369 mean IoU, while identifying precise spatial localization as the primary bottleneck for joint grounding success.

Tech stack: Python, LoRA (PEFT), GRPO, Transformers, PyTorch, TRL, Multimodal AI, Quantization, CVAT, HPC cluster, DDP, Grafana

Sep 2025 – Feb 2026
Wolfsburg, Germany

Volkswagen AG

Applied AI Engineer Intern

- Architected a **Multi-agent (7) hierarchical system** and custom **MCP** servers with strict **auth** and **observability**, backed by scalable **FastAPI** microservices to orchestrate complex, multi-domain user workflows (**Sound, Color, Projection & Memory**).
- **Developed a multi-turn chatbot** with **dynamic agent orchestration**, enabling seamless **user profile creation**, **iterative feedback handling**, and **real-time personalization** workflows.
- Optimized system architecture to reduce **latency** by 20% and **minimized hallucinations**—validated via **human evaluation** and mitigated using an **LLM-as-a-judge** framework—achieving 90% accuracy in AI-driven personalized theme generation.

Tech stack: Python, LangChain, LangGraph, MCP, FastAPI, LLMs, Redis, LangFuse

Feb 2025 – Jul 2025
Karlsruhe, Germany

IPEK, KIT

AI Research Assistant

- Developed conversational AI systems to integrate user-defined inputs into an **LLM-based autonomous driving scenario generation pipeline** for simulation-ready testing environments.
- Engineered logical, abstract, and concrete driving scenarios using **LLMs, RAG pipelines, Vector Databases, and AI agents** to generate critical simulation scenarios.

Tech Stack: Python, LLMs, RAG, LangChain, AI Agents, Vector Databases, Prompt engineering, SQL, Chainlit, chatbot design

Aug 2024 – Sep 2025
Siegen, Germany

IPEM, Universität Siegen

AI Research Assistant

- Architected **reinforcement learning and process control pipelines** for **industrial manufacturing environments**, focusing on minimizing cycle times (round-trip time) and optimizing automated production cells.
- Developed **digital twin simulations** (Unity) and data pipelines to model real-time industrial processes, improving RL learning efficiency and enabling scalable process analytics.

Tech Stack: RL, Optuna, Unity, OpenAI Gym, Digital Twin Simulation, MQTT, Stable Baselines3

EDUCATION

Oct 2022 – Aug 2026 Siegen, Germany	Master of Science in Mechatronics <i>Universität Siegen</i> Machine learning, Machine Vision, Sensorics, Industrial Automation & ADAS.
Jun 2016 – Sep 2020 Hyderabad, India	Bachelor of Technology in Mechanical Engineering <i>Vidya Jyothi Institute of Technology</i> CAD, Structural analysis, Automobile engineering & Thermal engineering.

PROJECTS

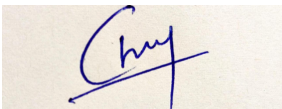
Enterprise Financial Analyst Agent <i>Agentic RAG System</i> Architected an enterprise-grade Agentic RAG system for DAX analysis, featuring a Dockerized ingestion pipeline using Docling for complex table extraction and a PostgreSQL/OpenSearch hybrid search (Vector + BM25) for high-fidelity retrieval across multilingual financial reports. Engineered a stateful agent via MCP with dynamic query decomposition and multi-turn memory, implementing a robust validation framework using RAGAS and a "Smart Degradation" protocol that intercepts missing data to prevent hallucinations by gracefully refusing to answer. Deployed a GDPR-compliant, LLM-agnostic containerized architecture (swappable across DeepSeek V4, NVIDIA Nemotron, Azure OpenAI, and local models) ensuring strict EU data residency and zero vendor lock-in, integrated with Langfuse for end-to-end observability and automated resilience via Tenacity. <i>Tech Stack:</i> LangGraph, NVIDIA Nemotron, DeepSeek, Gemini, PostgreSQL, OpenSearch, MCP, Docling, Docker, Streamlit, Langfuse, RAGAS, Tenacity, Python (Asyncio/SQLAlchemy)
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SKILLS

Programming Languages C, C++, Python, SQL, Object-Oriented Design (OOD), Data Structures & Algorithms	Software Tools CI/CD, Git, Docker, Microsoft 365, Web Frontends (Streamlit/Chainlit), Redis, FastAPI, LangFuse, MLFlow, A2A, Apache Airflow
AI/ML VLMs, Finetuning, Prompt engineering, Agentic AI, Reinforcement Learning (RL), vLLM, RAG, Reasoning & Planning, MLOps, LLMops, MCP, Vector databases, Conversational AI, Evaluation, Deployment	Libraries Numpy, Pandas, Scikit-learn, Matplotlib, PyTorch, OpenCV, Hugging Face, LangChain, LangGraph, Unsloth, TRL, RAGAS
Cloud Azure, AWS	Languages English - C1 , German - B1

INTERESTS & HOBBIES

Badminton <i>Regular active player in local club</i>
Trekking <i>Enthusiastic outdoor hiker</i>
Reading <i>Focused on technology trends, self-development, and non-fiction.</i>



Chinnaswamy Purra
Wolfsburg, den 28. September 2026